

=== GNB (1780339927) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth
To run the srsRAN Project gNB with this config, use the following command:

```
# sudo ./gnb -c gnb_zmq.yaml
```

```
cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1 # eMBB (3GPP TS 23.501, Table 5.15.2.2-1)
    inactivity_timer: 2700  # Sets the UE/PDU Session/DRB inactivity timer to 2700 seconds (45 minutes).

ru_sdr:
  device_driver: zmq        # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
  srate: 11.52             # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75              # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75              # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500         # ARFCN of the downlink carrier (center frequency).
  band: 3                  # The NR band.
  channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 15           # Subcarrier spacing in kHz used for data.
  plmn: "00101"           # PLMN broadcasted by the gNB.
  tac: 7                   # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0          # Set search space zero index to match srsUE capabilities
      coresets0_index: 6   # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common      # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64        # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log    # Path of the log file.
  all_level: info          # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false        # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: true        # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true        # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true        # Enable RLC metrics
```

```
enable_sched: true           # Enable DU scheduler metrics
periodicity:
  du_report_period: 400      # Set DU statistics report period to 400ms
```

=== UE (1780339927) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 52
nof_prb = 52

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1780339927) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#
# A copy of the GNU Affero General Public License can be found in
# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#

services:
  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    environment:
      - MONGODB_IP=127.0.0.1
      - OPEN5GS_IP=10.53.1.2
      - UE_IP_BASE=10.45.0
      - UPF_ADVERTISE_IP=10.53.1.2
      - DEBUG=false
      - SUBSCRIBER_DB=001010000000001,00aabbccddeeff112233445566778899,opc,112233445566778899aabbccddeeff00,8000,9,10.
      - NETWORK_NAME_FULL=srsRAN
      - NETWORK_NAME_SHORT=srsRAN
      - TZ=Europe/Madrid
    privileged: true
    ports:
      - "9898:9999/tcp"
    # Uncomment port to use the 5gc from outside the docker network
    # - "38412:38412/sctp"
    # - "2152:2152/udp"
    command: 5gc -c open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

  gnb:
    container_name: srsran_gnb
    # Build info
    image: srsran/gnb
    build:
      context: ..
      dockerfile: docker/Dockerfile
      args:
        OS_VERSION: "24.04"
```

```

# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  name:
    ipv4_address: ${GNB_IP:-10.53.1.1}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  content: |
    # This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
    # This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.
    # To run the srsRAN Project gNB with this config, use the following command:
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      bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
      supported_tracking_areas:
        - tac: 7
          plmn_list:
            - plmn: "00101"
              tai_slice_support_list:
                - sst: 1 # eMBB (3GPP TS 23.501, Table 5.15.2.2-1)
            inactivity_timer: 2700 # Sets the UE/PDU Session/DRB inactivity timer to 2700 seconds (45 minutes)

    ru_sdr:
      device_driver: zmq          # The RF driver name.
      device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_rate=11.52e6 # Optionally pass arguments to the RF driver.
      sr_rate: 11.52             # RF sample rate might need to be adjusted according to selected bandwidth.
      tx_gain: 75                # Transmit gain of the RF might need to be adjusted to the given situation.
      rx_gain: 75                # Receive gain of the RF might need to be adjusted to the given situation.

  cell_cfg:
    dl_arfcn: 368500            # ARFCN of the downlink carrier (center frequency).
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    plmn: "00101"              # PLMN broadcasted by the gNB.

```

```

tac: 7                                # Tracking area code (needs to match the core configuration).
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  common:
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  dedicated:
    ss2_type: common                  # Search Space type, has to be set to common
    dci_format_0_1_and_1_1: false    # Set correct DCI format (fallback)
prach:
  prach_config_index: 1               # Sets PRACH config to match what is expected by srsUE
pdsch:
  mcs_table: qam64                   # Sets PDSCH MCS to 64 QAM
pusch:
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log:
  filename: /tmp/gnb.log              # Path of the log file.
  all_level: info                     # Logging level applied to all layers.
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  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true                 # Enable RLC metrics
    enable_sched: true               # Enable DU scheduler metrics
  periodicity:
    du_report_period: 400            # Set DU statistics report period to 400ms
gnb_compose_config.yml:
content: |
cu_cp:
  amf:
    addr: ${OPEN5GS_IP:-10.53.1.2}
    bind_addr: 0.0.0.0
metrics:
  autostart_stdout_metrics: true
  enable_json: true
remote_control:
  bind_addr: 0.0.0.0
  enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
metrics:
  ipam:
    driver: default
    config:
      - subnet: 172.19.1.0/24

```

=== NOTAS (1780339927) ===

1. ****PLMN Coherence (Rule 1):**** MCC 001, MNC 01, TAC 7 applied consistently across gNB, UE (IMSI), and Open5GS (SUBSCRIBER_DB).
2. ****Frequency Coherence (Rule 2):**** dl_arfcn 368500 is valid for Band 3.
3. ****SCS Coherence (Rule 3):**** common_scs 15 kHz used.
4. ****ZMQ Ports (Rule 4):**** Ports 2000 and 2001 are crossed between gNB and UE.
5. ****bind_addr (Rule 5):**** gNB config has `bind\_addr: 10.53.1.1`, but `gnb\_compose\_config.yml` overrides it to `0.0.0.0`.
6. ****ru_sdr (Rule 6):**** Block copied exactly from CAG, ensuring 11.52e6 sr/s.
7. ****IMSI/SUBSCRIBER_DB (Rule 7):**** IMSI, K, OPC, AMF, SQN, and static IP are correctly formatted and matched between gNB and UE.
8. ****SST Value:**** SST 1 for eMBB extracted from RAG (ejemplos_configuraciones_5g.pdf_p141__chunk21) and commented with `#`.
9. ****UE PRB:**** `max\_nof\_prb` and `nof\_prb` set to 52 for 10 MHz bandwidth, 15 kHz SCS.